

Design tip

In a recent speech in Austin, Kay Hofmeester and Daniel Wigdor, both user experience experts and members of the Microsoft Surface team, commented that input is a language—one which must be created by the designer, learned by the user and taught with a new type of user interface. So whether it is based on touch, gesture or speech, a user interface must also include the necessary features to teach the user how to use it properly. This learning should hopefully be smooth and natural. A good technology with a sharp learning curve will be more of a horror than a pleasure!

brought it along from office. The concept of Sparsh is similar, where the user appears to have become a USB drive but the magic is happening in the cloud," Mistry adds.

Bend or twist, for a change

One of the software-based factors in improving user-interfaces is the zooming user interface (ZUI). Although the concept has been around for over a decade, the vision of creating the best ZUI continues to fire many an innovator.

Basically, a ZUI is a type of graphical user interface in which the users can change the scale of the viewed area in order to see more detail or less, and browse through different documents. Instead of constraining the user's view with fixed-size windows, the ZUI assumes the screen to be an infinitely-sized one, with information elements placed along the way. The user can pan the screen, zoom in or zoom out, and generally view things as they are placed in their mind.

The ZUI concept has powered the imagination of many device-makers including Sony, Apple, Google and Microsoft, leading to constant innovations. Every company has quite a few ZUI experiments dotting its timeline. If you take Microsoft, you will find that DeepFish, Canvas for OneNote and some Kinect applications are all examples of its trysts with ZUIs. ZUIs form the basis of successful tools like Google Maps and Microsoft Photosynth.

But the present ZUIs are nothing compared to what they will really be once we have flexible displays. In November 2011, Nokia demonstrated a prototype handheld device that lets the user bend and twist the screen to complete actions like scrolling and

zooming. Samsung too will soon be debuting phones with flexible displays, followed by tablets.

Such flexible displays will be the ultimate ZUI technology. You will have durable, flexible and large displays. And, you would just have to bend the screen or twist it to zoom in or see along the sides. Or, perhaps you can fold it to focus on some areas and ignore the rest. You can roll up your screen and put it into your pocket while you travel. Once you reach a coffee shop, you can spread the screen atop your table and have fun!

Or, chant a spell!

Voice training is a problem with voice-controlled devices because it is very difficult to achieve the noise-free environment or constant voice tone and accent required by these nascent tools. A clogged nose can play havoc, causing all your instructions to be misunderstood by the systems.

A lot of research is happening across the world to make speech system training more natural and accurate. Several toolkits and developmental aids are also coming up to enable

researchers, designers and developers. One recent success is the Open Source Kaldi toolkit.

Kaldi is written in C++ and provides a speech recognition system based on finite-state transducers, using the freely available OpenFst, together with detailed documentation and scripts for building complete recognition systems. It is more full-fledged than other ASR toolkits like HTK, Julius (both written in C), Sphinx-4 (written in Java) and RWTH ASR (written in C++). It works on common platforms like UNIX, Linux and Windows.

The advent of Siri instils a little more confidence in voice recognition. Apple's brainchild, Siri on the iPhone 4S allows you to handle text, reminders, Web search, etc using voice. Siri and other new-generation technologies like Nuance are a bit more accurate than older voice recognition software. Plus, with a little bit of intelligence, Siri is often able to understand in just one or two attempts what you are trying to say. Quite like a personal assistant, you can actually tell Siri to "call the cook and tell her to make pasta for dinner." You can dictate a whole e-mail or ask the phone to recommend restaurants nearby, all by simply talking to it.

Siri is the outcome of Apple's research partnerships with organisations ranging from DARPA to Bing, Yahoo and Google. As a result, the natural language user interface is quite mature and intelligent.

Other voice recognition tools include Google's Voice Actions and Microsoft's TellMe speech interface. However, these are not really speech application platforms but simply programs that process your voice and carry out some instruction. There is no high-level integration.

Real speech interface

For a long time now, developers have been hoping for a whole speech platform that not only allows you to interact with specific built-in functions but is also extensible to third-party applications just like a real operating sys-

Common goals of user-interface designers

1. Make the technology or equipment easy and enjoyable to use
2. Minimise user effort in understanding and using the device
3. Make the interaction more human-like
4. Reduce unnecessary interactions—be it input or output
5. Avoid clutter; keep it simple
6. Improve portability and energy-efficiency
7. Focus on health and environmental factors
8. Cater to users or applications with special needs